Data cleaning and description

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I - Loading and describing the data

a. Residential property market data

https://www.data.gouv.fr/datasets/indicateurs-immobiliers-par-commune-et-par-annee-prix-et-volumes-sur-la-periode-2014-2024

From this website, we downloaded 11 csv files, each file corresponding to a specific year from 2014 to 2024. Each file gathers aggregate variables related to residential property market at the municipality scale.

The data have been collected by the government, from 2014 to 2024. The latest update was on the 7th of July 2025.

i. Methodology

Data come from the treatment of another datagouv database regarding the Request for Property Values. From this initial database, they took only single-sale transfers, for which the values lied between 15,000€ and 10,000,000€, the surface between 10m² and 250m² for apartments, and between 10m² and 400m² for houses.

The population is the residential property transactions in Mainland France. The transactions have been aggregated at the municipalities level, which means that each line corresponds to a municipality for a unique year. Each municipality is identified by its INSEE code.

It is important to note that this database covers the whole of metropolitan France, with the exception of the departments of Bas-Rhin, Haut-Rhin, and Moselle, where the relevant data fall under the *livre foncier* (land registry) system.

One of the main strengths of this database is that it provides a common reference framework over a near-national geographical scope, making it possible to compare territories with one another, to understand their specific characteristics, and to produce studies and indicators based on homogeneous methods.

ii. Data cleaning and merging

After standardizing the labels, we merge our data using bind_rows in order to get panel data.

iii. Data descriptions

Type	Number
Columns	10
Total Rows	320598
Observations 2014	27691
Observations 2015	28201
Observations 2016	28385
Observations 2017	26687
Observations 2018	27074
Observations 2019	30231
Observations 2020	30470
Observations 2021	31101
Observations 2022	30915
Observations 2023	30011
Observations 2024	29832

Variables:

- INSEE code: a qualitative variable which is unique for each municipality
- Year
- The number of transfers: quantitative variable which is the sum of the house sales and flat sales
- The number of house sales: quantitative variable
- The number of flat sales: quantitative variable
- The proportion of house sales in percentage: quantitative variable
- The proportion of flat sales in percentage: quantitative variable
- The average price of properties sold in €: quantitative variable
- The average price per square metre of properties sold in €: quantitative variable
- The average surface area of properties sold in square metre: quantitative variable

b. Population evolution and stucture by municipalities

https://www.insee.fr/fr/statistiques/8581696?sommaire=8581933

i. Methodology

From this website, we downloaded csv file corresponding to “Évolution et structure de la population en 2022 - Commune - France hors Mayotte”

We have population statistics in 2011, 2016 and 2022 for each municipality in France, except Mayotte, corresponding to the January 2025 geography.

They have been collected by the INSEE during the census.

The data cover population distribution by sex, age, occupations but also housing conditions and relocation.

ii. Data cleaning and merging

Labels

When we downloaded the data sets, we had two csv files, one with the data we want to look at, in which the labels had a specific code, and the other with the labels' code and their meaning. Then, we changed the code labels of our data set of interest to get their meaning.

Structure of the dataset

Initially, we had one row for each municipality and the variables for each year. For example, we have 3 columns related to the number of men in the municipality, one for 2022, 2016 and 2011 (“Pop Homme en 2022”, “Pop Homme en 2016”, “Pop Homme en 2011”). Therefore, in order to get a panel data structure as the DVF data set we uploaded in the first part, we used at first pivot longer and we extract the year in each initial label. Then we create an additional column “Annee” and we used pivot wider to have the panel structure.

iii. Data descriptions

Type	Number
Columns	116
Total Rows	104709
Observations 2011	34903
Observations 2016	34903
Observations 2022	34903

Variables:

- Population: quantitative variable - number of residents in each municipality
- Per age range population (0-14, 15-29, 30-44, 45-59, 60-74, 75-89, 90+)
- Men population
- Per age range men population (0-14, 15-29, 30-44, 45-59, 60-74, 75-89, 90+)
- Women population
- Per age range women population (0-14, 15-29, 30-44, 45-59, 60-74, 75-89, 90+)
- Number of residents who live in the municipality for more than 1 year and who live in the same accommodation than 1 year ago
- Number of residents who live in the municipality for more than 1 year and who live in a different accommodation than 1 year ago
- Number of residents who lived outside the municipality 1 year ago (in a different department, in a different region, in Mainland France or in DOM) and their distribution by age and by sex.
- Number of residents in each CSP for every age range and sex.

c. Unemployment

https://www.observatoire-des-territoires.gouv.fr/taux-de-chomage-des-15-64-ans-rp

i. Methodology

From this website, we downloaded the file corresponding to "Taux de chômage des 15-64 ans", which was initially in the xlsx format. We therefore opened it on Excel and stored it as a csv file.

The data have been collected by the Insee, using the census in 2011, 2012, 2016 and 2022.

The data set gather, for each municipality and for each year of census, the total number of unemployed residents, the female unemployed residents and the male unemployed residents.

ii. Data cleaning and merging

na_chomeur	obs
104625	104625

Initially, for each year, each municipality had three rows:

- Total number of unemployed residents

- The number of female unemployed residents

- The number of male unemployed residents

To get a consistent panel structure, we created additional columns, that specify to what each number refers, and we used pivot wider.

iii. Data description

Type	Number
Columns	6
Total Rows	104625
Observations 2011	34875
Observations 2016	34875
Observations 2022	34875

Variables:

- INSEE_COM: as the previous data bases, this one uses the Insee code to classify the municipality
- libgeo: it corresponds to the name of the municipality associated with the INSEE code
- Annee
- Sex: coded as a factor, it expresses the gender category of the observation, distinguishing between females, males, and the total population
- Chomeur: number of unemployed people (based on the INSEE's definition of unemployment)

d. Merging all the databases

From the 3 data bases, we only kept data from 2016 and 2022, years of census and common to all bases. Using left join, because we wanted to keep all the observations we have in the initial panel set and match them with additional data we have from the two other data sets, we merged the data sets. Before merging, we ensured that the structure of the three data sets where coherent to be merged. We did this work in previous sections.

e. Department labels data

Source: https://www.data.gouv.fr/datasets/departements-de-france

i. Methodology

This file gather: - The department code - The department name - The region code - The region name

We use it here to add to our Merge data set the name of the department to which each municipality belongs. This will be useful latter to do some data visualization at the department level.

f. Geographical data

https://geoservices.ign.fr/adminexpress

i. Methodology and data description

The ADMIN EXPRESS product line centralizes five related datasets: ADMIN EXPRESS, ADMIN EXPRESS COG, ADMIN EXPRESS COG CARTO, ADMIN EXPRESS COG CARTO Petite Échelle, and ADMIN EXPRESS COG CARTOPLUS PE. Together, these datasets provide detailed administrative boundary layers for France across multiple levels. They include the following geographic units:

- Arrondissements (departmental and municipal)
- Cantons
- Administrative seats (chef-lieux) at various levels: arrondissement, municipal arrondissement, EPCI, canton, territorial collectivity, commune, associated or delegated commune, département, and region
- Communes and associated or delegated communes
- Départements
- EPCIs (public inter-municipal cooperation bodies)
- Regions

The main product, ADMIN EXPRESS, provides large-scale administrative boundaries and is updated monthly. An additional edition, ADMIN EXPRESS COG, is aligned with the Code Officiel Géographique (COG) published annually by INSEE.

This product line facilitates combining administrative boundaries with external datasets to build thematic representations of the French territory at different administrative levels. For our project, we mainly rely on the commune layer, as it gives the spatial unit necessary to merge geographic boundaries with demographic, employment, and real-estate datasets at the municipal level.

Here, we import both layers using `st_read()`, then standardize their INSEE municipality codes by trimming extra spaces and ensuring they always have five digits. This harmonization is essential for correctly joining the geographic layers with our DVF database, which also uses INSEE codes as identifiers.

```
Reading layer `chef_lieu_de_commune' from data source
  `/Users/lounaduyck/Documents/M1/BARBARIT-DUYCK-Project/Data/ADE_4-0_GPKG_LAMB93_FXX-ED2025-
  using driver `GPKG'
Simple feature collection with 34740 features and 9 fields
Geometry type: POINT
Dimension:     XY
Bounding box:  xmin: 102330 ymin: 6051834 xmax: 1239156 ymax: 7108510
Projected CRS: RGF93 v1 / Lambert-93

Reading layer `commune' from data source
  `/Users/lounaduyck/Documents/M1/BARBARIT-DUYCK-Project/Data/ADE_4-0_GPKG_LAMB93_FXX-ED2025-
  using driver `GPKG'
Simple feature collection with 34746 features and 16 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: 99041.5 ymin: 6046547 xmax: 1242424 ymax: 7110497
Projected CRS: RGF93 v1 / Lambert-93
```